

Final answers (record here — graded on the answer; show your work):

1. _____ 2. _____ 3. _____ 4. _____ 5. _____ 6. _____

Solving Quadratics by Factoring

Example (Solve by factoring):

$x^2 + 5x + 6 = 0$. Factor: $(x + 2)(x + 3) = 0$. Set each factor to 0: $x = -2$ or $x = -3$.

Directions: Factor, then use the Zero Product Property to solve.

1. $x^2 + 7x + 10 = 0$

4. $x^2 - 2x - 15 = 0$

2. $x^2 - x - 12 = 0$

5. $x^2 + 6x = 0$

3. $x^2 - 9 = 0$

Challenge Problems

6. $x^2 - 5x = 14$ — first make one side 0, then solve.

Answer Key and Procedure

Procedure

1. Move all terms to one side so it equals 0.
2. Factor the quadratic.
3. Set each factor = 0 and solve for x .

Answers

1. $x = -2, -5$
2. $x = 4, -3$
3. $x = 3, -3$
4. $x = 5, -3$
5. $x = 0, -6$
6. $x = 7, -2$